

Transient Thermal Characterization and Dimensionless Correlations for a Structured Silicon Carbide Volumetric Solar Receiver, with Consequences for Reduced-Order Model Validation

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Abstract

A monolithic silicon carbide honeycomb receiver — 100 square channels of 1.5 mm side, 0.40 mm webs, 137 mm long, 0.623 open frontal fraction — was characterized in a high-flux solar simulator at three lamp configurations and air flows of 4.5 to 18.3 sL min⁻¹, giving nominal channel Reynolds numbers of 23 to 94. Fifteen heating and three cooling transients were reduced to dimensionless form without recourse to a thermal model, and the receiver's constants identified from the transient eigenstructure rather than by inverse simulation. Gas–solid exchange is limited at the assembly scale rather than in the channel: the apparent Nusselt number, referenced to the length-averaged wall temperature, scales as $Re_{nom}^{1.472 \pm 0.011}$, one to two orders of magnitude below the fully developed laminar square-duct values, and the slow thermal eigenvalue identifies an effective capacitance of $275 \pm 38 \text{ J K}^{-1}$, six times that of the bare monolith.

Two results contradict standard practice in receiver modelling. First, the transfer-unit count increases with flow: obtained by integrating the measured axial wall profile, it scales as $Re_{nom}^{+0.341 \pm 0.040}$, where a single-stream formulation whose volumetric conductance is any fixed function of axial position requires $Re_{nom}^{-1.00}$. Flow-independent axial conductance profiles fitted instead — shared across runs, shared as Nusselt profiles, and fitted within each irradiance configuration, with up to seven free non-negative nodes — leave outlet residuals of 36 to 68 K root-mean-square that are monotone in flow, so no tested family reproduces the measured response. Second, the same fifteen runs yield a correlation prefactor spanning a factor of 29, an inversion threshold spanning 0.57 to 0.79 and a capacitance spanning 122 to 281 J K⁻¹, according only to which solid thermocouple is nominated as the reference and which probe subset and fit window are used. A model validated against a gas outlet temperature and one or two solid temperatures is therefore matched to a quantity that is not a property of the receiver, which offers a concrete explanation for the field's long-standing observation that measured absorber performance falls short of prediction while calibrated models appear to fit. We identify the instrumentation that would convert these installed-receiver response indices into physical coefficients.

Keywords: volumetric solar receiver; silicon carbide honeycomb; local thermal non-equilibrium; transient identification; model validation; dimensionless correlation

1. Introduction

Volumetric solar receivers, one route to convert solar energy to thermal or chemical energy, distribute concentrated radiation into the depth of a porous or structured absorber so the solid temperature peak sits inside the material rather than on the irradiated face, suppressing reradiation from the hot front surface and raising attainable outlet temperature. State of the art analyses anticipate thermal efficiencies above 75% with air outlet temperatures beyond 1000 °C, silicon carbide absorbers usable to 1500 °C where metallic meshes are confined to 800–1000 °C [1]. Three decades of prototypes have delivered unevenly: the wire-mesh TSA absorber reached 85% efficiency at 700 °C outlet and the recrystallized SiC SOLAIR absorber 70–75% at 750 °C [3,6], but Sulzer 1 delivered 68% against 80% predicted by its own design code, and Catrec 2 was destroyed attempting to raise its 460 °C outlet temperature [1,6]. A recent review by He et al. [46] states that the theoretical volumetric effect has not been validated by experiment, and most tested absorbers perform worse than predicted, continuing what Ávila-Marín et al. reported earlier as well [3].

The experimentalists' response has largely been to build better absorbers: foams and fibre meshes characterized for thermophysical and heat transfer properties [4,5], and various designs such as the twenty-six graded-porosity wire meshes and five SiC morphologies including Voronoi structures [6,7], single- against multi-layer ceramic foams [8], and pore-diameter variations at fixed porosity [9]. Current work continues the pattern — a 5 kW reticulated-porous-ceramic cavity receiver compared across SiSiC, alumina and ceria [31]; screen-printed ceramic and metallic-foil three-dimensionally shaped absorbers in a sixteen-module matrix at DLR's Synlight facility, reaching 735 °C exit and 91.8% efficiency at 650 °C against 84.6–85.3% for the 80 cpsi HiTRec SiSiC benchmark [29]; and a calorimetric bench measuring heat transfer to pressurised gas across rectangular channel geometries [32]. The recurring output is a design recommendation — porosity, cell density, layering, channel aspect ratio — from steady-state efficiency measurements. Supplementary section S1 offers a few more campaign-by-campaign details.

In parallel, the reduced-order modelling literature treats the receiver most commonly as a simple porous continuum or a single channel, despite the complexities, with coupled local thermal non-equilibrium and radiation models. Kribus et al. [2] show an idealized absorber in local thermal equilibrium achieving above 95% efficiency at 800 kW m⁻² and 1000 °C, unrealistically optimistic against the roughly 70% real absorbers achieve; Fend et al. [11] reproduce an extruded SiC honeycomb to within 15 K in outlet temperature and two efficiency points, with the extinction coefficient of 2300 m⁻¹ measured independently by transmittance, but with the volumetric heat transfer coefficient computed from a finer single-channel model rather than measured; and Ávila-Marín et al. [13] develop homogeneous equivalent models with a P1 radiation approximation and close the gas–solid coupling with volumetric correlations of the form $Nu_{lv} = a Re^b Pr^c$, fitting the respective coefficients. Superposing separately computed conduction and radiation contributions fails to recover the coupled effective conductivity [14], and a Rosseland treatment can differ from discrete ordinates by up to 76% [15]. Our own group [16] showed specular rather than diffuse channel reflectivity flattens the source term and raises modelled efficiency to 96% against 85% for the

commercial HiTRec-II geometry using a similarly simplified approach validated against a single data point. What these approaches share is that the gas–solid temperature difference is built into the formulation rather than observed, and how large this difference becomes is set by the volumetric coupling coefficient. However, that coefficient is taken from a correlation for another geometry, computed from a finer-scale model, or fitted to a few measured points. So the quantity that governs how the receiver behaves is an input to such models, not something any of them measures.

The channel-scale physics has been analysed more carefully outside the solar field, in the catalytic-monolith literature, and — importantly for what follows — with its validity conditions stated explicitly. Cornejo, Nikrityuk and Hayes [17] computed laminar convection in monolith channels under constant wall temperature and constant wall heat flux across Re of 50–600, showing that evaluating the Péclet number at the inlet rather than at the local temperature introduces errors of 50% or more, and that at high heating rates a minimum appears in the Nusselt–Graetz curve some 16% below the asymptotic value. Effective axial and radial conductivities are established for honeycomb substrates [18], and the multi-scale comprehensive studies note radiation is required to reproduce measured axial temperature profiles and that a thermocouple inside a monolith channel can be significantly affected by radiation even without blocking the channel [19]. A 0.05 mm wall gap substantially reduces effective radial conductivity [20], and a pseudo-continuous two-dimensional reduction reproduces full three-dimensional CFD to better than 0.5 K [21]. These results, however, rest on assumptions their authors state: flow equally distributed over all channels [21], and the gas treated as static [20]. Because the assumptions are stated, it is easy to check them, and they do not hold for an absorber under a concentrated, spilling beam with an uncontrolled flow split. The channel-scale transport those studies establish is not in question; its transfer to this case is.

Neither family of studies, experimental or computational, measures an installed receiver through its transients and reads the governing quantities directly off the measurement, which matters because calibration data for a reduced-order model are almost always a gas outlet temperature with one or two embedded solid temperatures. Capuano and co-workers [10] illustrate the consequence: reviewing four established DLR models against one reference experiment, they report predicted outlet air temperatures of 1050, 1048, 1057 and 1073 K against 972 K measured, and efficiencies of 0.78, 0.77, 0.79 and 0.81 against 0.69, attributing the gap partly to the difficulty of experimentally evaluating a mean air flow temperature. Four independently developed models agreeing with each other to within 25 K yet disagreeing with the experiment by 80–100 K signals a validation target that fails to constrain what the models disagree about.

This work, first, brings a new and extensive experimental dataset (in section 2) of eighteen transients — fifteen heating and three cooling — each recorded at nine axial and radial locations of a lab-scale receiver, for various gas flow rates and irradiance levels. Second, presents an analysis framework (in section 3) that reduces the measurements to dimensionless groups, identifies the receiver's constants from the transient eigenstructure, and tests the structure of reduced-order formulations against the measured exponent. Instead of an inverse calibration fitting a coefficient's value, it only explores which way the transfer-unit count moves as the flow rises. Any formulation whose volumetric conductance is a fixed function of axial position requires that count to fall in proportion to $1/\dot{m}$, whereas the measurement herein has it rising; a disagreement in sign cannot be absorbed by

recalibration, so the conclusion holds whatever coefficient is chosen. The framework also quantifies how much the identified coefficients depend on the researcher's choice of reference sensor, and thus how much information a conventional validation exercise carries. These yield (section 4) a dimensionless characterization of the installed SiC honeycomb receiver; the demonstration of the wall-to-interior disequilibrium based on the measurements rather than assumptions; identification of effective capacitance and loss conductance from transient eigenvalues with sensor-selection sensitivity quantified; a similarity rescaling on which the fifteen heating transients collapse. Finally, the work utilizes the experimental data and analysis framework to support two methodological findings (in section 5) — that validation against a gas outlet temperature plus one or two solid temperatures is largely uninformative for this class of receiver, and that a uniform-temperature or uniform-flux coefficient evaluated on the fixed channel geometry cannot reproduce its flow response at any calibration.

2. Experiment

2.1 Receiver and apparatus

The absorber is a monolithic silicon carbide honeycomb (Figure 1d) with a ten-by-ten array of square channels of 1.5 mm side and 0.40 mm web thickness, giving a 1.9 mm pitch, a 19.0 mm square frontal face of area 3.61 cm², an open frontal fraction of 0.6233 and an axial length of 137 mm. The measured mass is 40 g, which over the solid cross-section of 1.360×10^{-4} m² and the 137 mm length gives an effective bulk density of 2147 kg m⁻³, consistent with a recrystallized SiC of roughly one third internal porosity; this is the value used for every capacitance estimate below. The geometry is of the same family as the extruded SiC honeycombs and catalyst-carrier substrates characterized in the solar literature [4,11] and modelled in our earlier work [16], with a channel size between the two quoted literature samples (1.43 and 2.18 mm) and a comparable web thickness.

The monolith is enclosed in an aluminum cavity (142 mm in diameter, depicted in Figure 1b, and utilized earlier as a plain solar cavity [47]) held in place with multiple layers of alumina felt insulation (by Shine Refractories), and irradiated through a wider circular aperture by a seven-lamp high-flux solar simulator (Figure 1a), each lamp is a 6 kW xenon short-arc unit; the apparatus and its optical characterization (Figure 1c) are described in detail elsewhere [22,23]. The receiver is operated in suction: cavity air is drawn in through the irradiated face, passes through the monolith to the exit plenum, and leaves through a rear flange cooled by a closed water loop held at 25 °C, so that the gas reaching the metering line is near ambient. The drop across that flange is large and is measured — 15 to 442 K between the plenum probe and the station immediately downstream of it across the campaign — which is why no station in the return line is used as the inlet reference. Air is metered by four Aalborg GFC17 thermal mass-flow controllers (MFC) of nominal 0 to 5 sL min⁻¹ range, all carrying air, referenced to 21.1 °C and 1 atm, and the total flow is their sum. Four units were used in place of a single 0 to 20 sL min⁻¹ device because the manufacturer specification is $\pm 1.0\%$ of full scale over the whole 0 to 100% range with $\pm 0.5\%$ full-scale repeatability, so subdividing the range reduces the absolute error provided the unit errors are not fully correlated, and section 3.4 reports both correlation limits for that reason. Only one of the four MFCs carries a factory calibration for air; the

other three are calibrated for different gases, so their indicated flow is converted, which scales each unit's air-equivalent range by its own conversion factor and takes the widest to 5.72 sL min^{-1} . Rather than rely on the manual's conversion factors, each unit was calibrated in place against a bubble flowmeter, which replaces the factors with a directly measured characteristic and leaves a residual proportional to reading. Both terms are carried in the uncertainty model of section 3.4. Per-unit readings span 0.69 to 5.72 sL min^{-1} , from the smallest set point the controllers hold — 10% of nominal range, below which flow was not controllable — to full range, so neither term is negligible against the other. Because these are mass-flow devices reporting standard volumetric units, the mass flow follows from the reference density $\rho_{\text{std}} = 1.1996 \text{ kg m}^{-3}$ rather than from the density at the prevailing ambient temperature.

2.2 Instrumentation

Figure 2 shows the piping and instrumentation diagram (P&ID), mainly the nine sheathed thermocouples (sensors/probes) characterizing the measuring campaign: six of type N (TCDirect UK) and three of type K (Omega UK). Of which, three probes (type N, OD 1.5 mm, length 150 mm) contact the monolith side wall at axial positions of 11, 58 and 107 mm, denoted T8, T12 and T11 (TI-8, TI-12 and TI-11 in Figure 2). Two probes (OD 0.5 mm, length 500 mm) sit in the interior, flow-exposed region at 58 and 107 mm, denoted T9 and T10. One probe, T3 (type N, OD 1.5 mm, length 300 mm), sits in the exit plenum and reports the outlet gas temperature. One further probe, T2 (type K, OD 1.5 mm, length 150 mm), sits in the surrounding insulation and is used in section 4.6 to assess thermal stationarity; the wall-to-insulation drop $T_{12} - T_2$ is 310 to 770 K, so the insulation resistance dominates radial transfer. The remaining two probes, T15 and T16 (both type K) placed a few cm behind and beside the cavity, respectively, give the setup's mean ambient temperature, T_{amb} . They are the least drifting temperatures in the rig, warming by 1.6 and 1.8 K between the first two minutes of a run and its steady window against 3.3 K for the probe downstream of the gas cooler, so the inlet reference is taken from them rather than from any station in the return line. Substituting either probe alone for their mean moves every quantity identified below by less than a tenth of its Monte Carlo standard deviation. The distinction between the wall chain and the interior pair is critical for this work. Sections 4.3 and 5.1 show that the two report materially different temperatures at the same depth, and that which is taken to be the solid temperature changes every identified coefficient in this work. Interior probe readings in similar geometries are known to be perturbed by the radiation field [19], which is why we treat the reference-temperature choice as a systematic rather than a detail. Thermocouple tolerance is treated per IEC 60584 class 1, which sets the same limit for type N and type K — the greater of 1.5 K and $0.004|t|$ with t in $^{\circ}\text{C}$ — and which at the hottest front-probe reading of 1204 K is ± 3.7 K. Installation error for the three sheathed wall probes, comprising contact resistance against the monolith and radiative exchange with the cavity, is not quantified.

Differential pressure across the monolith is monitored with a Keller PD33X of ± 200 mbar range and $\pm 0.1\%$ full-scale accuracy (PI-1 in Figure 2, hereafter DP1), hence an accuracy floor of ± 0.2 mbar. Section 4.7 shows this to be the same order as the entire expected monolith pressure drop, so the sensors readings are not useful. Nonetheless, these are still presented because a corrective action has be devised since the quantity of pressure drop can assist on discriminating between the two mechanisms discussed in section 5.2.

2.3 Campaign and reduction

Fifteen heating runs span three lamp configurations at nominal aperture irradiances of 456, 304 and 256 kW m⁻²: runs E67–E71 at 7.1 to 15.3 sL min⁻¹, E72–E76 at 4.5 to 18.3 sL min⁻¹, and E77–E81 at 4.5 to 13.8 sL min⁻¹ respectively. Three cooling transients, C69, C80 and C81 at 10.5, 6.6 and 4.5 sL min⁻¹, follow lamp shutdown of the heating runs E69, E80 and E81 respectively, with flow maintained. Two additional runs, E82 and E83, repeat E77 and E81; they are retained in the dataset for repeatability check but excluded from every fit reported here. All raw data are available in the work's repository (see Data Availability section)

Steady-state values are 120 s means at the end of each heating run. The length-averaged wall temperature is the length average of the piecewise-linear profile through the three wall thermocouples with constant extrapolation to the end faces,

$$\bar{T}_w = 0.2518 T_8 + 0.3504 T_{12} + 0.3978 T_{11},$$

using the exact trapezoid coefficients for thermocouples at 11, 58 and 107 mm over 137 mm. Gas enthalpy differences are computed by trapezoidal integration of $c_p(T)$ between T_{amb} and the relevant gas temperature rather than as $c_p(\bar{T})\Delta T$. The measured conditions of the fifteen steady runs are collected in Table 1.

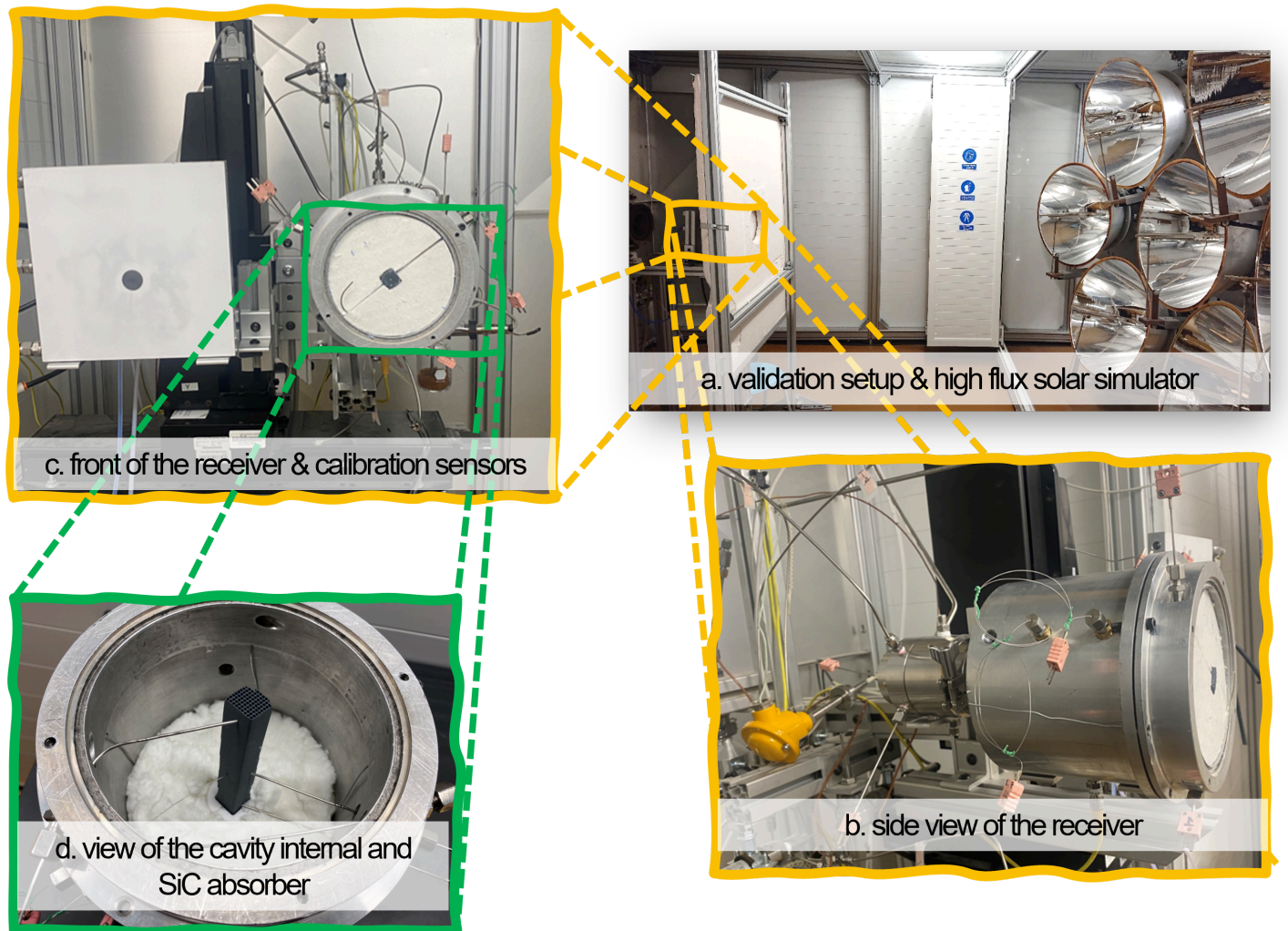


Figure 1. The experimental assembly: a) The seven-lamp high-flux solar simulator with the receiver mounted on its optical axis; b) Side view of the receiver, showing the cylindrical aluminium cavity and the thermocouple leads entering through its wall; c) Front of the receiver with the alumina-coated Lambertian target and the calibration sensors used for the flux mapping of section 4.6; d) View of the opened cavity (without most of the insulation layers), showing the silicon carbide honeycomb seated in alumina felt insulation. Sensor axial positions are given in Figure 2.

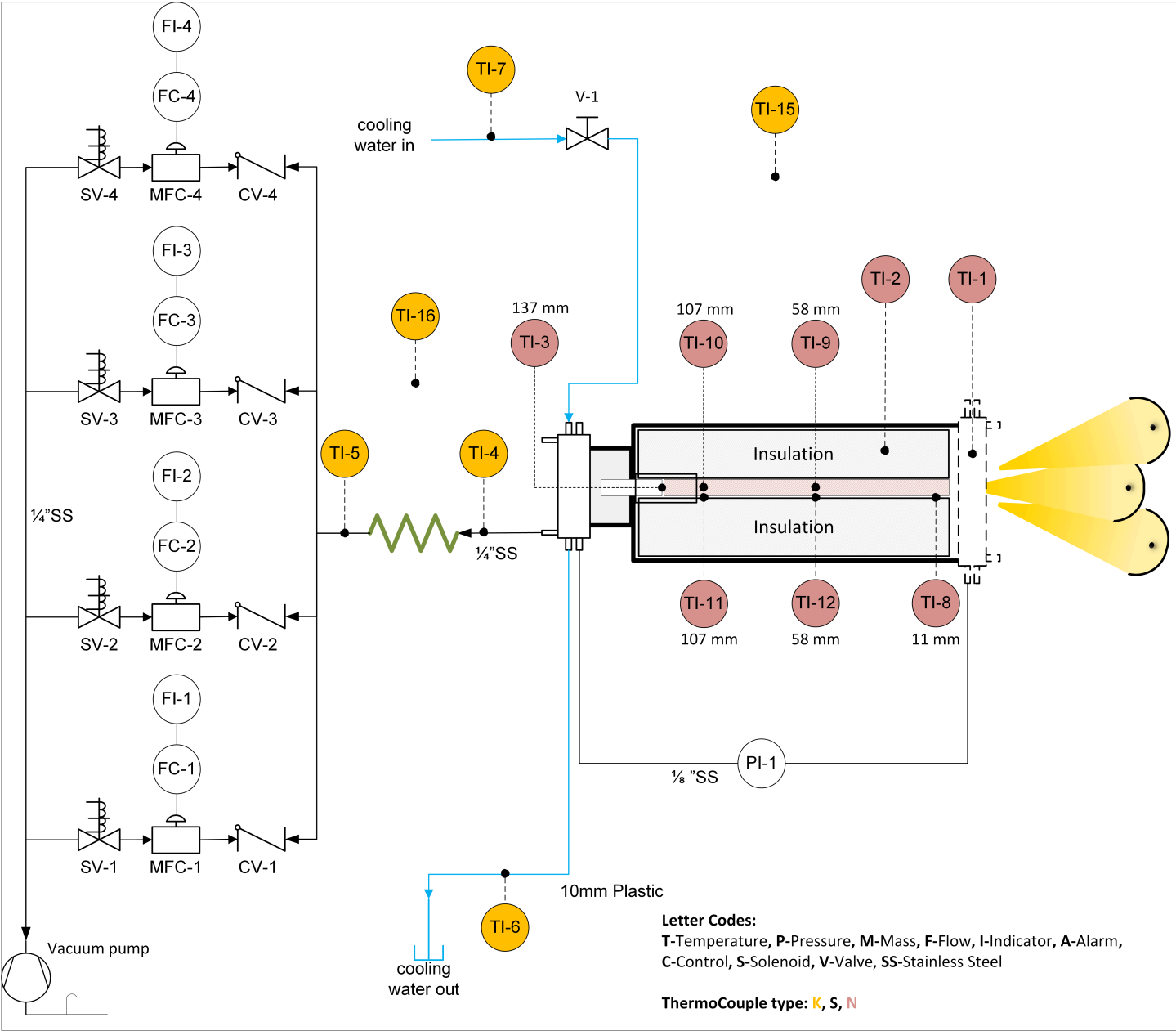


Figure 2. Piping and instrumentation diagram (P&ID), with the letter codes and thermocouple type colour codes given in the panel.

Run	$G_0[kWm^{-2}]$	$q[sLmin^{-1}]$	$T_{amb}[K]$	$\bar{T}_w[K]$	$T3[K]$	$T12 - T8[K]$
E81	256	4.53	297.8	700	550	-9.4

Run	$G_0[kWm^{-2}]$	$q[sLmin^{-1}]$	$T_{amb}[K]$	$\bar{T}_w[K]$	$T3[K]$	$T12 - T8[K]$
E80	256	6.61	296.6	686	561	+25.8
E79	256	8.04	296.7	669	560	+39.6
E78	256	10.01	298.1	643	554	+54.0
E77	256	13.85	297.1	589	525	+58.1
E76	304	4.53	296.4	908	647	-97.9
E75	304	6.95	296.7	911	681	-29.1
E74	304	9.03	296.5	896	697	+13.8
E73	304	13.17	296.9	840	694	+56.6
E72	304	18.32	296.4	757	653	+60.7
E71	456	7.13	300.8	1047	753	-83.5
E70	456	9.11	299.0	1045	779	-31.1
E69	456	10.50	296.7	999	770	+11.5
E68	456	12.50	298.7	979	773	+35.1
E67	456	15.28	296.3	935	764	+53.5

Table 1. The 15 heating conditions (measured quantities and fixed averages) of the campaign ordered by irradiance then flow rate.

3. Analysis framework

This chapter defines the quantities the campaign's experimental data is reduced to. All are computed directly from the measured records; no thermal or radiation model of the receiver is fitted to them. This approach yields steady state dimensionless groups treating the assembly as a single-stream heat exchanger (section 3.1), an effective capacitance and loss conductance identified from the transient decays (section 3.2), a rescaling that tests the lumped description those identifications presume (section 3.3), and the uncertainty accounting (section 3.4). Dependencies on additional, to the methodology framework assumptions, and other reference choices are declared in place. The chapter closes with the details implementation of the framework along with the necessary libraries and packages (section 3.5).

3.1 Dimensionless groups

Per-channel mass flow is $\dot{m}_{ch} = \dot{m}/100$ and the hydraulic diameter is the channel side, $D_h = 1.5$ mm. Gas properties are evaluated at $\bar{T}_g = (T_{amb} + T_3)/2$ except the thermal conductivity in the Nusselt number, which uses the film temperature $(\bar{T}_w + \bar{T}_g)/2$ — as discussed earlier, because evaluating the Péclet number on an inlet rather than a local temperature shifts the apparent Nusselt–Graetz curve by 50% or more [17].

The receiver is characterized as a single-stream heat exchanger following Kays and London [25], with effectiveness (ε) and transfer-unit count (NTU)

$$\varepsilon = \frac{T_3 - T_{amb}}{\bar{T}_w - T_{amb}}, \quad NTU = -\ln(1 - \varepsilon),$$

giving an apparent assembly-level coefficient h_{app} and Nusselt number:

$$h_{app} = \frac{NTU \dot{m}_{ch} c_p}{PL}, \quad Nu = \frac{h_{app} D_h}{k(T_{film})}.$$

Re is evaluated with $\dot{m}/100$, assuming equal division of metered flow among the hundred channels — an assumption sections 5.2 and 5.3 argue is violated — so a nominal Reynolds number, Re_{nom} , is required. Similarly, NTU_{app} , h_{app} and Nu_{app} are calculated algebraically from the outlet-plenum probe T3 and an axial wall average, in the absence of independently measured interfacial coefficients. All four are considered installed-receiver response indices, appropriate for the heat transfer analysis herein but not explicit transport coefficients and transferable to another geometry; extracting physical coefficients is left to a follow companion modelling study.

Two important notes. First, the ε –NTU relation is exact for a wall at fixed temperature, whereas here the wall spans up to 321 K between T8 at the front and T11 at the back, and the maximum sits in the interior: the axial profile inverts, with T12 above T8 in 10 of the 15 runs. The inversion's onset moves to higher flow as irradiance rises: the lowest flow at which it is observed is 6.6, 9.0 and 10.5 sL min^{−1} at 256, 304 and 456 kW m^{−2}, bracketing interpolated onsets of 5.1, 8.4 and 10.1 sL min^{−1} (Table 3). The bias is quantified in section 4.2. A side-wall inversion indicator $\Delta T_w = T_{12} - T_8$ is used throughout: negative while the front-face probe is the hotter, positive once mid-depth overtakes it. Second, since h_{app} is defined through NTU, $Nu(Re)$ is algebraically tied to $\varepsilon(Re)$:

$$\frac{d \ln Nu}{d \ln Re} = 1 + \frac{d \ln NTU}{d \ln Re} + \frac{d \ln [c_p/k]}{d \ln Re}.$$

Because c_p is evaluated at the bulk mean and k at the film temperature the ratio c_p/k is being used rather calling it a Prandtl number. A fitted $Nu(Re)$ exponent is therefore not independent corroboration of the measured $\varepsilon(Re)$; the flow dependence that carries information is that of the transfer-unit count, and section 5.2 makes the respective comparison.

The apparent wall-to-interior disequilibrium is expressed by the below Λ ratios, at each of the two receiver depths (58 and 107 mm) which are utilized as a surrogate of gas–solid nonequilibrium in section 4.3:

$$\Lambda_{58} = \frac{T_{12} - T_9}{T_{12} - T_{amb}}, \quad \Lambda_{107} = \frac{T_{11} - T_{10}}{T_{11} - T_{amb}}.$$

Finally, four dimensionless groups are used in section 5.2 to capture the physics: Biot number $Bi = h_{app}t_w/2k_{SiC}$, Graetz number $Gz_L = D_h Re Pr / L$, axial Péclet group $Re \cdot Pr \cdot L / D_h$, and radiation–conduction number $N_{rc} = 4\sigma\bar{T}_w^3 D_h / k(\bar{T}_w)$.

3.2 Transient identification

After lamp shutdown, during cooling runs, the assembly's temperature decays with a single slow mode. For each sensor (probe) the excess temperature over the second half of the record, and above 5 K, is regressed as $\ln(T - T_{amb})$ against time, and λ is the mean of the per-sensor slopes. Lumping the assembly's thermal behavior into one capacitance C_{eff} losing heat to ambient through a conductance K_{loss} while delivering enthalpy to the gas gives:

$$\lambda = \frac{\varepsilon \dot{m} c_p + K_{loss}}{C_{eff}}.$$

So regressing the obtained λ against the gas advective conductance $x = \varepsilon \dot{m} c_p$ allow identification of both constants from slope and intercept with no need for solving a more complex mathematical model. Note that, each cooling transient is paired with the heating run sharing its identifier (EXX/CXX) and takes that run's effectiveness.

Identifying a heat transfer property from a transient rather than from a steady balance is established practice for these materials with Fend et al. [4] showing how volumetric heat transfer coefficients of porous absorbers have been obtained by measuring how far a temperature wave is delayed and damped in passing through a sample. The same logic is applied here to the full assembly.

In principle the heating branch decays with the same eigenvalue, and $\ln(T_{ss} - T)$ is regressed against time in the same way. It differs in that the asymptote T_{ss} is estimated from the tail of the same record, which forces a choice of fit window. We use the middle of that record, $0.07 < u < 0.45$, where $u = (T_{ss} - T)/(T_{ss} - T_0)$ is the fraction of the total rise still to come - equal to 1 at lamp-on and 0 at steady state. Earlier than $u=0.45$ the faster modes have not yet decayed; later than $u=0.07$ the remaining deficit is comparable to the uncertainty in T_{ss} itself. The choice of the window is not arbitrary. Sweeping the window from (0.05, 0.35) to (0.20, 0.70) moves C_{eff} monotonically from 295 to 200 J K⁻¹ while r^2 improves from 0.84 to 0.97, so fit quality cannot select it (supplementary section S3). Furthermore, the temperature probe set alone changes C_{eff} by more than a factor of two (discussed in section 4.4).

We therefore report cooling as the primary identification, since its asymptote is measured and it needs no probe rule once the source is removed; heating remains as a conditional consistency check; and a joint fit to all eighteen eigenvalues offers an overall estimate. Their disagreement is addressed in section 5.1.

3.3 Similarity rescaling

If the lumped description holds, every temperature transient should collapse when time is rescaled on the identified eigenvalue as $t^* = t/\tau$ with a time constant of $\tau = C_{eff}/(\varepsilon \dot{m} c_p + K_{loss})$. The ε factor enters τ because it is the effectiveness that converts the metered flow into the conductance the

eigenvalue actually sees; both the reference constants and the effectiveness are taken from the primary identification and from each run's own measured effectiveness. The resulting half-rise is a property of the receiver's response shape rather than a quantity with a predicted value: only an ideal single-node system would give $\ln 2$, and the departure from $\ln 2$ measures how far the receiver is from a single state.

3.4 Uncertainty

The uncertainty of the calculated quantities is propagated through each measurement, based on the instruments' specification, and calculation step. Since all the framework is implemented over a computer script (see section 3.5) it is advantageous to utilize a Monte Carlo simulation [50] for uncertainty propagation and estimate it across many "realizations". One realization perturbs every measured input by a random draw (ie Monte-Carlo random selection) from its specified error and repeats the entire reduction on the perturbed record; after testing various counts, four thousand realizations give a representative and count-independent ensemble for each derived quantity, whose spread is the reported uncertainty. Because each physical error is drawn once and applied wherever it acts, correlations between derived quantities are carried by the ensemble rather than assumed. Three kinds of uncertainty are propagated and reported separately : i) Instrument error, ii) systematic bias, and iii) estimator choice

Instrument (random and calibration) error

For the thermocouples, the IEC 60584 class 1 type N and K tolerance — the greater of 1.5 K and $0.004|t|$ — is a maximum permissible error rather than a standard deviation, so it is read as a 95% coverage bound and is halved to a 1σ systematic term, drawn once per sensor per realization and shared across runs, with an independent 0.5 K drift term per run; viscosity and conductivity carry 2% and 1% perturbations.

The metered flow sums four controllers carrying roughly 25, 33, 28 and 14% of the total, and their error is two-term because the two available pieces of information constrain different things: the manufacturer specification is $\pm 1.0\%$ of full scale with $\pm 0.5\%$ full-scale repeatability, an additive bound in absolute flow, while the in-house bubble-flowmeter calibration constrains the slope of each unit's characteristic and so contributes a term proportional to reading. Each unit's deviation therefore is $\sigma = [(0.0025F)^2 + (0.025q_u)^2]^{1/2}$, drawn once per unit per realization and combined with each record's own measured shares. F is the unit's air-equivalent full scale, the nominal 5 sL min^{-1} scaled by its gas-conversion factor; instead of using the four individual factors we assign every unit the widest, $F = 5.72 \text{ sL min}^{-1}$, which over-states the additive term and is therefore conservative. The two terms (manufacturer specification and bubble-flowmeter calibration) have opposite flow dependence, so the relative uncertainty on the summed flow falls from 2.8% at the lowest campaign flow to 2.5% at the highest for fully correlated units, and from 1.4% to 1.3% for independent ones. The two limits agree to within 0.0001 on every exponent and 0.5 J K^{-1} on every capacitance, so which of them is used is immaterial; hereafter the larger is quoted with the flow error being predominantly systematic. All fitted-slope quantities are accompanied by the regression standard error. For the pooled correlation and the transfer-unit count of section 4.2 that standard error exceeds the instrumental term by an order of magnitude and therefore this should be used as the uncertainty; for the better

determined grouped correlation it exceeds the instrumental term only 3.3-fold. Supplementary section S2 details the propagation.

Systematic Bias

Systematic bias in the outlet gas thermocouple is propagated with the selected range being much larger than the calibration error. T3, in the exit plenum, sits at a convective–radiative equilibrium between gas and hardware and reads neither bulk gas nor wall temperature; the bias is potentially of order tens of kelvin and uncharacterized — a principal source of the model–experiment gap in this field [10], demonstrated for thermocouples in monolith channels specifically [19]. T3 enters ε , NTU, h_{app} , Nu, ε^* and η directly. It reaches C_{eff} and K_{loss} only through the effectiveness, which forms the abscissa $x = \varepsilon m^* c_p$ of the eigenvalue regression. Each therefore carries a $\delta T3 = \pm 25$ K band alongside its Monte Carlo interval, evaluated on the transient identification as well as on the steady groups. Section 4.4 evaluates it and finds the identified capacitance to be the quantity least sensitive to the outlet-thermocouple systematic. Its impact is further discussed in section 5.2.

Estimator choice

Where more than one reduction paths (referred as estimator choice) exists, as for the inversion threshold ε^* , their individual uncertainty propagation is estimated. Specifically, section 4.1 reports ε^* as estimated both by interpolation between bracketing runs and by global linear regression, considering this a systematic difference potentially affecting the analysis conclusions.

3.5 Implementation

The reduction and figure generation are implemented as two Julia 1.12.6 scripts (available, for details check Data Availability and Reproducibility) executed in the repository environment with `julia --project=. ,` with dependencies fixed by `Project.toml` and `Manifest.toml`. `receiver_reduction.jl` reads the raw logger records and generates all reduced data, regressions, uncertainty estimates, tables and the JSON results archive using CSV 0.10.16 and DataFrames 1.8.2 for data handling, GLM 1.9.5 for linear regression, Roots 3.0.4 for scalar inversions, Optim 2.2.1 with ForwardDiff 1.4.1 for the multi-start conductance-profile fits, Interpolations 0.16.3 and Trapz 2.0.3 for property interpolation and enthalpy integration, and JSON3 1.14.3 for serialization. `make_figures.jl` imports the shared reduction definitions without rerunning the complete reduction, regenerates plotting traces from the raw records when required, and renders Figures 3–7 using PythonPlot 1.0.6 with Matplotlib 3.9.1. Dry-air properties at 1 atm are evaluated using CoolProp 8.0.0 through the Julia `CoolProp_jl1` binary: the Helmholtz equation of state of Lemmon and co-workers supplies (c_p) [48], while the Lemmon and Jacobsen correlations supply viscosity and thermal conductivity [49], consistent with the formulations used by REFPROP. Because `PropsSI` is a scalar call whereas the two 4000-realization Monte Carlo cases evaluate these properties several million times, CoolProp is queried once when the script is loaded on a 1 K grid spanning 200–1600 K, after which the call sites use linear interpolation. The interpolation error remains below (10^{-6}) relative over 250–1500 K, and the grid brackets the campaign range of approximately 294–1200 K, so no experimental evaluation requires extrapolation. The Monte Carlo propagation and multi-start optimizer use fixed random seeds; consequently,

rerunning the scripts with the archived Julia environment reproduces the outputs, apart from possible last-digit differences between numerical-library builds.

4. Results

The reduced quantities for every run, referred to throughout this section, are given in Table 2; the measured conditions they derive from are in Table 1.

Run	Re_{nom}	Gz_L	ε	NTU_{app}	N_{prof}	Nu_{app}	Λ_{58}	Λ_{107}	η_{nom}
E81	25.1	0.192	0.628	0.988	1.065	0.0380	0.0472	0.0661	0.252
E80	36.3	0.277	0.680	1.138	1.213	0.0642	0.0505	0.0751	0.385
E79	44.2	0.338	0.707	1.227	1.296	0.0853	0.0528	0.0817	0.466
E78	55.3	0.422	0.742	1.355	1.408	0.1197	0.0576	0.0907	0.565
E77	78.5	0.600	0.781	1.518	1.540	0.1951	0.0635	0.1108	0.694
E76	23.2	0.178	0.573	0.851	0.925	0.0282	0.0343	0.0547	0.297
E75	34.7	0.266	0.625	0.982	1.058	0.0494	0.0374	0.0636	0.501
E74	44.6	0.341	0.669	1.106	1.179	0.0727	0.0404	0.0714	0.681
E73	65.2	0.499	0.731	1.314	1.369	0.1302	0.0455	0.0900	0.985
E72	93.6	0.715	0.773	1.481	1.512	0.2162	0.0531	0.1144	1.223
E71	33.8	0.259	0.607	0.933	1.011	0.0444	0.0306	0.0608	0.407
E70	42.5	0.326	0.644	1.032	1.109	0.0627	0.0322	0.0664	0.554
E69	49.4	0.378	0.674	1.120	1.192	0.0802	0.0340	0.0733	0.628
E68	58.6	0.449	0.697	1.192	1.257	0.1027	0.0357	0.0812	0.750
E67	72.2	0.553	0.731	1.315	1.369	0.1417	0.0374	0.0939	0.902

Table 2. The per-run quantities that follow from the definitions of section 3 (reduction framework). Runs are ordered by irradiance then flow rate, as in Table 1.

4.1 Steady field, flow response, and the inversion criterion

The steady temperatures against flow at each configuration are shown in Figure 3, and the response is strongly graded in depth. Front-face wall temperature falls steeply with flow, at -33.2 , -24.3 and -20.8 K per sL min^{-1} at 456, 304 and 256 kW m^{-2} with r^2 of at least 0.97; the mid-depth wall falls at half to two-thirds that rate; the rear wall barely falls, at -1.1 , -1.9 and -5.2 K per sL min^{-1} with r^2 of 0.07, 0.12 and 0.76. Outlet gas temperature is nearly flow-invariant at the two higher fluxes ($+0.6$ and

−0.1 K per sL min^{−1}, $r^2 \leq 0.03$) and falls only weakly at the lowest, at −3.1 K per sL min^{−1}. At 456 kW m^{−2} the outlet gas spans 26 K across the flow range against 264 K for the front-face wall, a ratio of ten; at 304 and 256 kW m^{−2} the spans are 51 K against 331 K and 36 K against 194 K respectively.

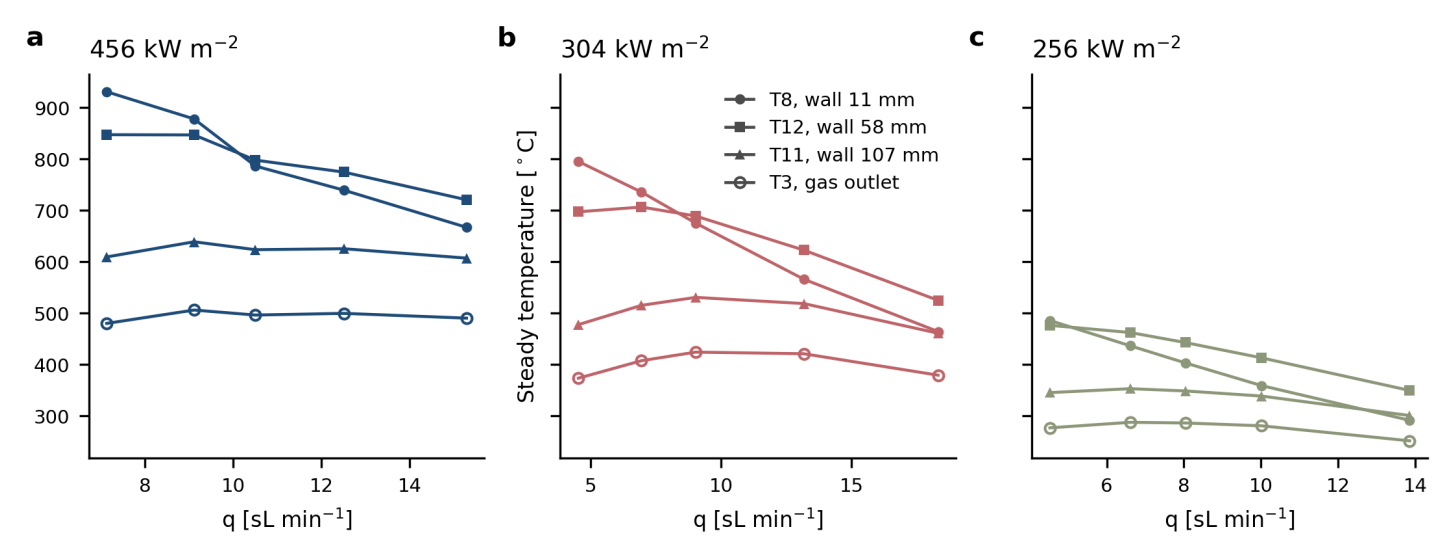


Figure 3. Steady and outlet gas temperatures against flow rate at the three lamp configurations.

Figure 5 presents the analysis of the axial temperature profile and, in particular, its inversion; Table 3 gives the condition at which the inversion occurs, expressed both as a flow rate and as an effectiveness ϵ . The side-wall profile inverts as flow increases, crossing zero within the tested range at all three configurations (Figure 5a), and the crossings collapse when expressed in effectiveness rather than flow (Figure 5b). However, there are two aspects to be considered before the threshold is read as a universal number. First, the inversion indicator is the difference between two side-wall probes, at 11 and 58 mm, so what it locates is a front-to-mid crossing along the receiver's axis, but measured on the wall. The true temperature maximum is three-dimensional and need not lie on that wall, since the light beam is Gaussian and carries radial gradients; the radial traverse proposed in section 5.3 would locate it. Second, the inversion threshold also depends on how the effectiveness is formed: ϵ^* uses the length-averaged wall temperature, and section 5.1 shows that its value changes when the reference probe changes. ϵ^* is therefore an operating criterion for this receiver and its instrumentation rather than a property of the receiver.

Nominal G_0 [kW m ^{−2}]	q^* local [sL min ^{−1}]	ϵ^* local	q^* global	ϵ^* global
456	10.12	0.666 ± 0.002	11.08	0.673
304	8.36	0.655 ± 0.002	10.32	0.673
256	5.08	0.642 ± 0.003	3.69	0.628

Table 3. Located inversion crossings at the three configurations, by local interpolation between the bracketing runs and by global linear regression, with the effectiveness at crossing.

While the two estimators agree to within 0.02 in ϵ^* (Table 3), they disagree enough to matter. At 256 kW m^{−2} the global regression puts the crossing at 3.69 sL min^{−1}, 18% below the lowest flow run at that

flux: only one of the five points in that group is below zero, and the indicator rises steeply from it before flattening, so a single straight line is too shallow at the bottom and carries the crossing outside the data. Local bracketing between the two runs that straddle zero keeps all three crossings inside the measured range and halves the ϵ^* spread, from 0.045 to 0.024; therefore it is the estimator adopted here.

The data support a crossing at $\epsilon^* \approx 0.65$, rising weakly from 0.642 to 0.666 as the flux rises 1.8-fold. The experimental parameter space is not sufficient to claim either flux independence or a resolved flux dependence. Although the 0.024 spread of effectiveness is well above the 0.002–0.003 Monte Carlo intervals (Table 3), it is still no larger than the difference between the two estimators. In any case, the form of the criterion is robust because the inversion is set by how much of the available wall-to-ambient temperature difference the gas has recovered, not by flow rate or flux separately. Its value, however, depends on the outlet probe: under the $\delta T_3 = \pm 25$ K band, ϵ^* at 256 kW m^{-2} runs from 0.579 to 0.704.

4.2 Assembly-scale heat exchange behavior

As per Shah and London [24] the fully developed laminar square-duct reference depends on thermal boundary condition and in their work they offer three standardized cases for unity aspect ratio: $Nu_T = 2.976$ for a uniform wall temperature, $Nu_{H2} = 3.091$ for a uniform wall heat flux both axially and peripherally, and $Nu_{H1} = 3.608$ for a uniform axial flux with an isothermal perimeter, with the product $f \cdot Re = 14.227$ throughout. The two flux conditions differ only in whether the wall conducts heat around the channel perimeter. We adopt Nu_{H2} as primary reference, since interior channels see similar radiative and convective conditions on all four walls and thin webs make a peripherally uniform flux the better idealization than a uniform temperature - in conjunction with the negligible web Biot number reported later. Nonetheless and irrespective of the reference case, the measured Nu_{app} (see Table 2) is a factor of 14.3 to 110 below Nu_{H2} , comparably far below channel-level Nusselt numbers of order three to four computed for monolith channels under either boundary condition [17]. These point to a rather limited heat transfer exchange at the full assembly scale and not necessarily at the channel (film) level.

Over the full campaign (Figure 4a), the apparent Nusselt number follows a clean power law , $Nu_{app} = (3.14 \pm 0.11) \times 10^{-4} Re_{nom}^{1.444}$, $r^2 = 0.968$, $n = 15$, with pooled (all data together) regression standard error ± 0.069 and instrumental Monte Carlo term ± 0.003 ; the regression term exceeds the instrumental one roughly twentyfold, so residual dispersion is physical and not driven by metrology or random errors. Fitted separately per irradiance configuration the exponent is 1.440, 1.476 and 1.530 at 256, 304 and 456 kW m^{-2} ($r^2 \geq 0.9996$ each), so pooled scatter reflects a flux-ordered offset plus a weak monotonic trend in the exponent. A common-slope model with a separate prefactor per irradiance configuration gives exponent 1.472, regression SE ± 0.011 , instrumental term ± 0.003 , $r^2 = 0.9993$, and prefactors 3.24 , 2.72 and 2.55×10^{-4} in the same order. This grouped model is reported as the primary correlation alongside the pooled fit, because it is the better determined of the two and because the data were acquired at three discrete irradiance levels rather than over a continuous sweep.

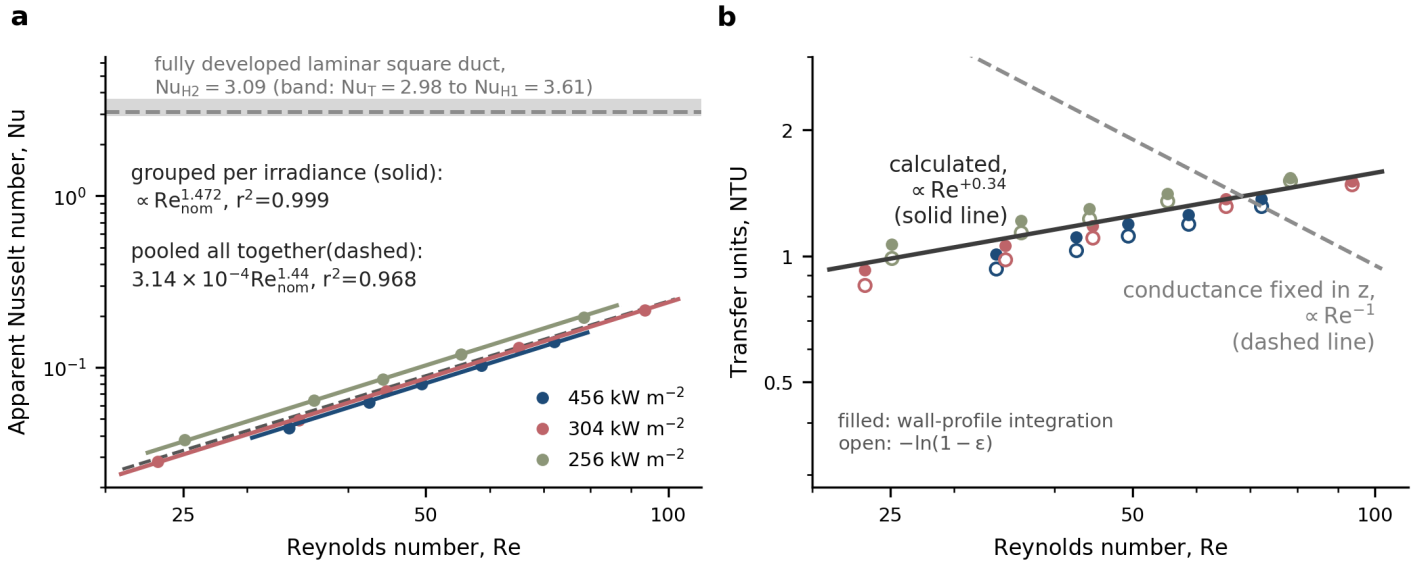


Figure 4. a) Apparent Nusselt number against nominal channel Reynolds number, with the fully developed laminar square-duct literature reference $Nu_{H2} = 3.09$ and the band spanning the other two thermal boundary conditions, $Nu_T = 2.98$ to $Nu_{H1} = 3.61$ [24]. Solid lines are the primary common-slope grouping per irradiance level; the dashed line is the pooled (all together) single-fit. b) Profile-corrected transfer-unit count against Reynolds number (filled symbols) with the isothermal-wall identity $-\ln(1 - \epsilon)$ (open symbols), and the respective slope fit on Reynolds number (solid line), against the Re^{-1} requirement of a conductance that is a fixed function of axial position (dashed line).

As section 3.1 established, the Nusselt exponent is the effectiveness trend with a definitional unit added to it, so the quantity carrying independent information is the transfer-unit count — a dimensionless measure of how much heat exchange the gas actually receives over the monolith length. Integrating the gas energy equation through the measured piecewise-linear wall profile with a uniform h , and solving for the $N = hPL/\dot{m}c_p$ that reproduces each run's measured outlet temperature, gives (Figure 4b)

$N_{prof} \propto Re_{nom}^{+0.341}$, ± 0.004 (instrumental), ± 0.040 (regression SE), $r^2 = 0.845$,
 These counts exceed the isothermal-wall identity values by 1.5% to 8.7%, largest at low flow where the axial wall gradient is steepest, and the identity itself gives $NTU_{app} \propto Re_{nom}^{(+0.389 \pm 0.045)}$. The wall temperature profile is not available over the full length because the probes start at 11 mm and finish at 107 mm, in other words there are no data for the front 11 mm and rear 30 mm of the monolith. Therefore, the exponent's validity to fully reconstruct the temperature profile is investigated over six plausible extrapolations (utilizing three temperature slopes: zero/ temperature equal with the neighboring probe, linear change from the neighboring probe, half the linear slope). SM section S4 presents the analysis with four out of the six extrapolations yielding admissible profiles. In particular, continuing the rear gradient at half slope moves the exponent only to +0.331, and back-extrapolating the front gradient moves it by less than 0.003. A full linear continuation of the rear gradient is inadmissible, since for five of the fifteen runs it places the exit wall temperature below the measured outlet gas temperature, which no single-stream model can produce — the outlet measurement therefore bounds how steep the rear wall temperature may decrease. All the admissible extrapolations

move the exponent at the ± 0.01 level, two orders of magnitude below the disagreement established next.

Transfer units increase with flow, and that direction is unavailable to an entire class of models. For a single gas stream exchanging heat with a solid whose volumetric conductance is any fixed function of axial position — ie $h(z)$ independent of mass flow — the transfer-unit count is given by,

$$N = \frac{1}{\dot{m}c_p} \int_0^L h(z) P dz \propto \dot{m}^{-1} \propto Re^{-1}$$

with no adjustable constant. The physical consequence of this is that a conductance fixed in position (ie only as function of length) has a fixed total over the monolith length, so raising the mass flow gives each unit of gas heat capacity a smaller share of it, and the exchanger must become less effective rather than more. Evaluated on the measured properties the requirement is -1.000 for a fixed Nusselt number and -1.016 for a conductance independent of temperature and flow, the two differing only through the temperature dependence of the gas properties; we write it as -1.00 below. The measured exponent is $+0.341$: a gap of 1.34 in the power of Re , and a disagreement in sign. The sign is what makes the result structural, since a mismatch in magnitude can be absorbed by recalibrating the coefficient, whereas no flow-independent $h(z)$ — however large, however distributed along the channel — can turn a count that must fall into one that rises. Neither is it an artifact of the outlet probe: under a uniform $\delta T_3 = \pm 25$ K offset the exponent moves only from $+0.320$ to $+0.377$, staying positive and over eight standard errors from zero, let alone from -1.00 . This is one of main contributions of this work and further elaborated in section 5.2.

The dimensionless groups (estimated as per section 3.1) validate the transport phenomena of the herein cases, beyond Re_{nom} and Gz_L that are shown in Table 2 the other values of the other auxiliary groups per run are tabulated in the SM section 5. The web Biot number never exceeds 3.4×10^{-5} , so the solid is isothermal across the web thickness, consistent with the high effective conductivities reported for honeycomb substrates [18] and with SiC at the conductive end of the substrates compared by Cui and Kær [20]. The Graetz number reaches only 0.704; its reciprocal, $x^* = 1/Gz_L$, runs from 1.42 at the highest flow to 5.71 at the lowest, against a fully developed asymptote reached at x^* of order 10^{-1} [24], so the channel is thermally fully developed over all requiring only a few percent of its length at every condition. The axial Péclet group is at least 1460, so axial gas conduction is negligible, and the radiation–conduction number is 1.50 to 5.66, so the solid redistributes absorbed energy axially by radiation — consistent with radiation being required to reproduce measured axial temperature profiles in monolith channels [19]. There are more interpretations that can be extended from these groups and therefore section 5.2 continues on the flow response.

4.3 Local nonequilibrium

Figure 5c presents the wall-to-interior disequilibrium at the two instrumented depths (58 and 107 mm), each normalised on its own wall-to-ambient difference. Both disequilibrium indices rise linearly with Reynolds number within every flux group and both are small in magnitude, Λ_{58} spanning 0.031 to 0.064 and Λ_{107} spanning 0.055 to 0.114. At 107 mm the three per-flux slopes agree to within $\pm 2\%$, with values of 8.77, 8.55 and 8.38×10^{-4} per unit Re at 456, 304 and 256 kW m^{-2} and r^2 of 0.997, 0.999 and 1.000. However, offsets are ordered by flux: $\Lambda_{107} = c(G_0) + (8.54 \pm 0.36) \times 10^{-4} Re$, a flux-invariant

slope with a flux-dependent offset falling as flux rises, intercepts 0.0440, 0.0342 and 0.0313 at 256, 304 and 456 kW m^{-2} . A pooled single-intercept fit averages the offsets away, gives $r^2 = 0.897$ against the 0.997–1.000 achieved within groups, and inflates the slope's standard error ninefold. At 58 mm the same structure holds at roughly a third of the slope — 1.81, 2.66 and 3.12×10^{-4} per unit Re, r^2 of 0.984, 0.999 and 0.992 — with Λ_{58} rising by 22%, 55% and 35% across the flow range at the three fluxes. Therefore, disequilibrium is a smooth monotonic function of depth and flow without demonstrating any threshold or hysteresis.

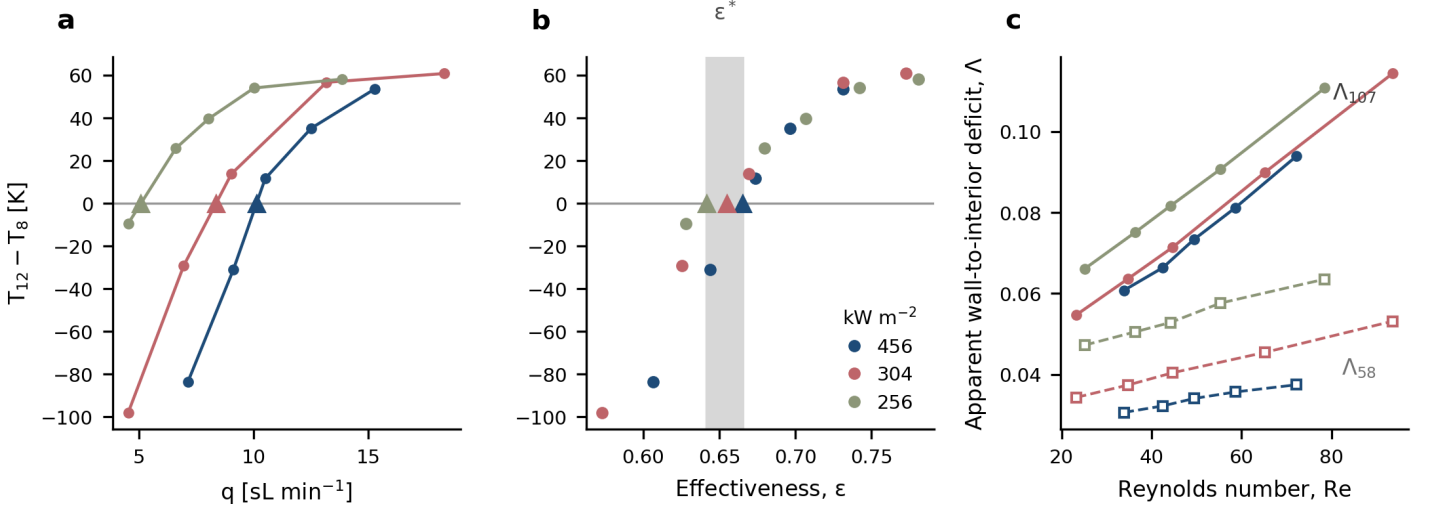


Figure 5. a) The side-wall inversion indicator $\Delta T_w = T_{12} - T_8$ against flow at the three configurations, with the located crossings. b) The same data against effectiveness, collapsing at $\epsilon^* \approx 0.65$. c) The wall-to-interior disequilibrium indices at 58 and 107 mm against Reynolds number, rising linearly within every flux group.

The T9 and T10 sheathed thermocouples are exposed to the gas flow, each expressing a measure of the bulk-gas measurements. However, they sit at a convective–radiative–conductive equilibrium among gas, surrounding solid and their own (long) stems, and for this geometry a probe inside a monolith channel is known to be significantly perturbed by radiation even unblocked [19]. Λ is thus an apparent wall-to-interior disequilibrium rather than a true gas–solid nonequilibrium, nonetheless a surrogate best considered as an upper bound, since radiative loading biases the interior probe toward the solid and understates the true gas–solid gap.

Local thermal non-equilibrium is built into the two-equation porous-absorber formulations [2,11,13,15] and of heterogeneous monolith models [18,21], its magnitude following from an assumed or fitted volumetric coefficient. A directly instrumented disequilibrium at two depths gives that literature something to check against, provided the comparison uses the same observation model rather than bulk-gas temperature. Extending the discussion in section 4.2 this gap grows with Reynolds number because the wall-to-interior probe difference widens as flow rises, yet the integrated transfer-unit count also rises. Taken at face value both point to a participating gas–solid contact that increases with flow; the conditional is discharged in section 5.2.

4.4 Transient identification and the energy budget

Figure 6 presents the transient analysis from which the receiver's thermal capacitance and loss conductance are identified, and Table 4 collects the resulting constants. The three cooling transients decay as a single exponential in all six receiver probes, with a per-sensor eigenvalue spread of 3.7% (Figure 6a). The abscissa $x = \varepsilon \dot{m} c_p$ (the gas-side advective conductance) requires an effectiveness for each cooling run; referring each run to the heating state it decays from, with c_p on that run's own tail temperatures, gives the primary identification (Figure 6b),

$$C_{\text{eff}} = 275 \pm 38 \text{ J K}^{-1}, \quad K_{\text{loss}} = 0.080 \pm 0.023 \text{ W K}^{-1}, \quad r^2 = 0.986.$$

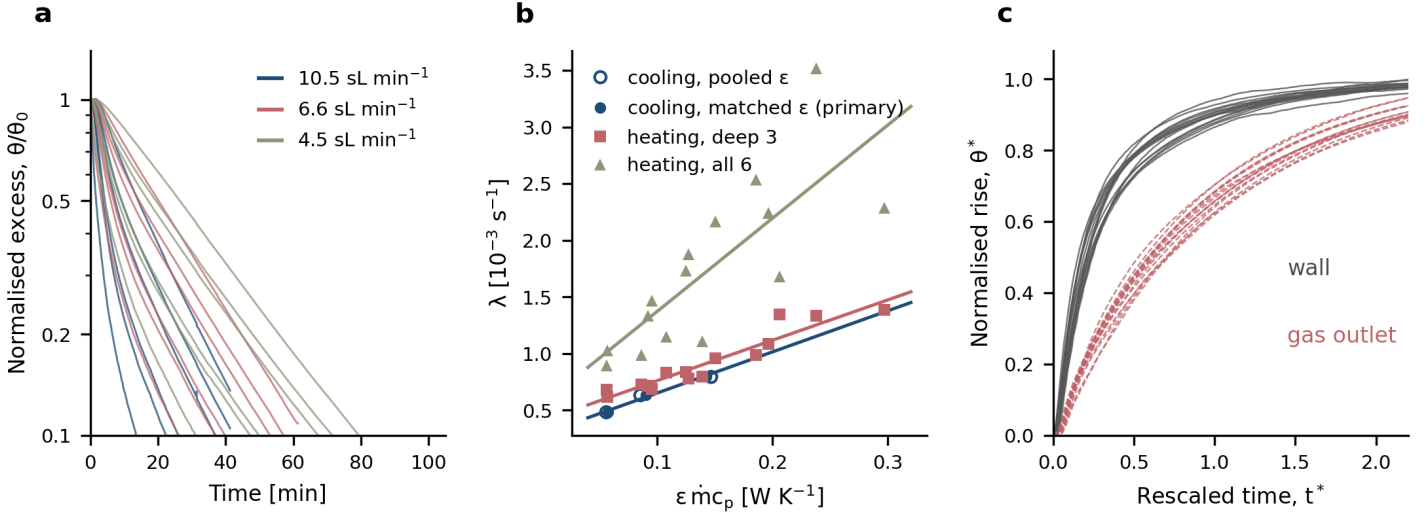


Figure 6. a) Cooling decays of all six receiver probes at three flows, normalised on the initial excess. b) The slow eigenvalue against gas advective conductance $\varepsilon \dot{m} c_p$ for the three identifications, the primary matched-effectiveness cooling points (filled symbols) and the pooled-effectiveness abscissa (open symbols). c) Similarity collapse of the fifteen heating transients on $\tau = C_{\text{eff}}/(\varepsilon \dot{m} c_p + K_{\text{loss}})$, wall (in grey) and outlet gas (in red).

A pooled $\varepsilon(q)$ correlation across all fifteen heating runs, averaging over the flux ordering of ε , gives $298 \pm 42 \text{ J K}^{-1}$ and $0.095 \pm 0.025 \text{ W K}^{-1}$ at $r^2 = 0.964$; the matched form is better-conditioned and 8% lower in capacitance. However, both rest on three eigenvalues and hence one degree of freedom. The joint fit to all eighteen eigenvalues gives $269 \pm 30 \text{ J K}^{-1}$ and $0.101 \pm 0.022 \text{ W K}^{-1}$ with sixteen degrees of freedom and $r^2 = 0.899$; heating-only identification on the deep and outlet probes gives $281 \pm 35 \text{ J K}^{-1}$ and $0.114 \pm 0.027 \text{ W K}^{-1}$. The spread, 269 to 298 J K⁻¹ (11%), is a measure of how well the dataset determines the capacitance (Figure 6b). The Monte Carlo mean of the matched capacitance, 280 J K⁻¹, sits above the point estimate because C_{eff} is the reciprocal of a slope fitted to three points.

Whichever estimate is taken, C_{eff} exceeds the bare monolith capacitance of 42.0 to 46.8 J K⁻¹ (from the measured 40 g mass and c_p at 600–900 K) by a factor of six to seven. Therefore, on this lumped description roughly five sixths of the thermal mass governing the receiver's transient is attributable to the holder and insulation rather than to the absorber. The attribution is by subtraction: the absorber's own capacitance is measured and the rest is the remainder. Unfortunately, the identification cannot resolve how that remainder divides between holder, insulation and the plenum hardware. The

installed capacitance governs the transient behavior of the assembly ie start-up and cycling estimates, but is invisible to steady-state characterization. This is one reason it is absent from a literature built on steady efficiency measurements [4,5,6,7,8,9], and it cannot be recovered from absorber material properties alone. It also qualifies He, Du and Shen's claim that porous volumetric receivers have quick transient response and small thermal inertia owing to the short flow path of the heat transfer fluid [46]. This is true for the absorber alone (42–47 J K⁻¹ is small), but not for the installed receiver, where the greater part of the responding mass belongs to hardware the absorber's properties say nothing about.

Transient behaviour has typically been characterized via a receiver time constant from outlet air temperature rather than by identifying the underlying capacitance. The same review of He, Du, and Shen reports time constants of about 90 s (Sulzer), 70 s (HiTRec II), 365 s (Sandia foam), 660 s (CeramTec) and 600–840 s (SOLAIR), noting no standard exists [46]. A time constant, $C_{eff}/(x + K_{loss})$, measured at one flow leaves numerator and denominator unseparated; the 70–840 s spread reflects that conflation. In particular, separation here comes from varying advective conductance, attributing the six- to sevenfold excess specifically to holder and insulation. Other studies utilized computational analysis. For example, a coupled transient model for start-up, shut-down, clear-sky and cloud passage [41], and ray tracing with pore-scale simulation and PID control giving 7.0 s (air) versus 45.9 s (molten salt), rising 129% as mass flow falls threefold [42]. A predicted time constant implies a predicted capacitance. Conversely, the factor found in this study cannot be captured unless holder and insulation are modelled explicitly.

Table 4 collects these constants with their Monte Carlo intervals. The loss conductance is bracketed between 0.080 and 0.114 W K⁻¹, with the heating tangent exceeding the matched cooling secant by 1.41× — consistent with a partly radiative loss path, though only weak evidence for T³ dominance (the cubic temperature dependence expected of a radiative loss mechanism). Carried through the $\delta T_3 = \pm 25$ K band, the primary determination moves over 271.7 to 280.5 J K⁻¹ and 0.0754 to 0.0844 W K⁻¹, and the heating determination over 262.2 to 299.4 J K⁻¹ and 0.1073 to 0.1200 W K⁻¹. The band widens the loss bracket to 0.075 to 0.120 W K⁻¹ while leaving the capacitance spread dominated by estimator choice (pair/set of thermocouples) with reasonably close values.

Constant	Value	s.d.	95% interval	Unit	Notes
Nu_{app} prefactor a (pooled)	3.14×10^{-4}	0.11×10^{-4}	[2.93, 3.37]	$\times 10^{-4}$	15 steady runs
Nu_{app} exponent, pooled	1.444	0.003	[1.438, 1.451]	—	instrumental MC; regression SE ± 0.072 , $r^2=0.968$
Nu_{app} exponent, grouped (primary)	1.472	0.003	[1.466, 1.479]	—	instrumental MC; regression SE ± 0.011 , $r^2=0.9993$

Constant	Value	s.d.	95% interval	Unit	Notes
N_{prof} exponent (primary)	+0.341	0.003	[0.335, 0.347]	–	instrumental MC; regression SE ± 0.040 ; fixed- Nu requirement -1.000
NTU_{app} exponent (identity, superseded)	+0.389	—	—	–	isothermal-wall identity; retained for comparison only
Inversion marker ε^* @456 kW m ⁻²	0.666	0.002	[0.661, 0.670]	–	operational marker under the adopted wall convention; see §5.1
@ 304 kW m ⁻²	0.655	0.003	[0.650, 0.661]	–	same
@ 256 kW m ⁻²	0.642	0.003	[0.635, 0.648]	–	same
Λ_{107} slope [Re^{-1}]	8.54×10^{-4}	0.36×10^{-4}	[7.85, 9.27]	$\times 10^{-4}$	common slope, per-flux intercepts
C_{eff} , cooling matched- ε (primary)	275	38	[219, 369]	J K ⁻¹	$n=3$, $r^2=0.986$
cooling pooled- ε	298	42	[237, 402]	J K ⁻¹	$r^2=0.964$
joint 18 eigenvalues	269	30	[221, 337]	J K ⁻¹	$r^2=0.899$
heating deep probes	281	35	[226, 361]	J K ⁻¹	122 J K ⁻¹ if all six probes used
K_{loss} , cooling matched- ε (primary)	0.080	0.023	[0.045, 0.136]	W K ⁻¹	secant conductance
heating deep probes	0.114	0.027	[0.071, 0.177]	W K ⁻¹	tangent conductance
Monolith capacitance (measured mass)	42.0 – 46.8	—	—	J K ⁻¹	$40 \text{ g} \times c_p(600\text{--}900 \text{ K})$

Table 4. Identified constants with Monte Carlo standard deviations, 95% percentile intervals and, for fitted slopes, the regression standard error alongside.

4.5 Similarity collapse

Rescaling time on $\tau = C_{eff}/(\varepsilon\dot{m}c_p + K_{loss})$, the lumped thermal time constant combining the receiver's effective heat capacity with the heat-input and loss rates, collapses all fifteen heating transients (Figure 6c). The length-averaged wall temperature reaches half its total rise at $t^* = 0.205$, where t^* is time normalised by τ , with a coefficient of variation (the run-to-run spread relative to the mean) of 16.8% across runs spanning a fourfold flow range and a 1.8-fold flux range. The outlet gas reaches half rise at $t^* = 0.634$ with a coefficient of variation of 7.8%. A single slow mode therefore is able to reflect the late heating transient, and the offset between the two half-rise values quantifies the phase lag between solid and gas. However, this does not establish a one-state response. The wall half-rise sits well below $\ln 2$, the dimensionless half-rise value expected for a single exponential mode, while the outlet sits close to it. So the wall retains a faster component that the lumped model does not represent, consistent with section 3.2's finding that the front probes carry their own short time constant. This can be attributed to the heating transients sharing a common flow-dependent time scale, set by the slow eigenvalue, on which their late behaviour collapses. A contradiction with most literature studies omitting or ignoring the dynamic behavior of the system, which in the case of receiver could be approximated with a one-parameter dynamic system, despite the startup/shutdown and control being the practical challenges of pilots and large scale applications.

4.6 Delivered aperture irradiance

The aperture irradiance is determined two ways, and they disagree. The first is radiometric flux mapping by the three-step characterisation procedure established for this simulator [22,23]. The procedure combines an alumina-coated Lambertian target on a three-axis slider, a Gardon-type circular foil radiometer traversed on a graded grid with NIST-traceable calibration and a calorimetric correction for spectral mismatch, and a parallel high-dynamic-range CCD reconstruction. Gaussian fits validate the focal distribution to a normalised RMSE, the root-mean-square deviation of the fit scaled by the peak signal, of about 0.07 and 0.12 on the two axes [22]. Integrating the calibrated map over the receiver face gives the nominal values 456, 304 and 256 kW m⁻² that label the three configurations.

The second is an apparent steady energy closure, which is not model-free. It takes K_{loss} from section 4.4 and omits storage by assuming the assembly is stationary at run end. Neither assumption is safe: the insulation probe rises from 308 to 356 K across the campaign, with a decay far slower than any run window. At steady state

$$f = \frac{Q_{gas} + K_{loss}(\bar{T}_w - T_{amb})}{G_0 A_{frt}}$$

should equal unity if the radiometric flux is correct. Here f denotes the ratio of the energy-balance estimate of aperture power to its nominal radiometric value. Evaluated at both ends of the K_{loss} bracket and averaged per configuration, it does not. f runs 0.989–1.134 at 456 kW m⁻², 1.149–1.324 at 304 and 0.784–0.916 at 256. Apparent efficiency on the nominal basis ranges 0.252–1.224, exceeding unity in the 304 kW m⁻² group. We therefore report spans of 451–517, 304–402 and 201–256 kW m⁻², and every power-normalised quantity as a band across them, not as a bounding interval. The closure

omits storage, inherits a lumped loss model and carries the T3 bias, an offset in the outlet-temperature reading, directly into Q_{gas} , so neither determination is an established bound.

Supplementary section S6 shows that no uniform outlet-probe shift and no flux-only correction reconciles the two, because the required correction is flow-dependent. That part of the gap at the two higher fluxes is structural: a Gardon gauge sensing over 499 mm^2 is unable to resolve the flux that a 361 mm^2 face intercepts in a focused spot, with a face-to-gauge ratio of 1.04 to 1.28 depending on spot width. The same difficulty of converting incident onto delivered power has been reported for this simulator elsewhere [23]. The residual requires a flow-dependent term in the energy budget; peripheral bypass and channel-group maldistribution are two candidates consistent with section 4.2. However, there are two aspects to be considered before treating this as confirmed. However, non-stationarity and a flow-dependent probe response are not excluded, so the convergence with section 5.2 is a consistency and not an independent confirmation.

No temperature-based result depends on the choice. Effectiveness, transfer units, the Nusselt correlation, the inversion criterion, the disequilibrium indices, the eigenvalue identification and the similarity collapse are all ratios of measured temperatures and flows. Finally, only the efficiency and delivered-power-per-unit-mass figures shift, and those are reported as spans rather than as intervals containing the true value.

4.7 Pressure drop

The installed differential-pressure measurement is assessed against the laminar square-duct prediction, to establish whether it resolves the monolith's own pressure drop. Steady-window means of the differential-pressure channel compare poorly with that prediction, computed with a Poiseuille number of 56.91 (the dimensionless friction factor-Reynolds number product that characterises fully developed laminar flow resistance in a square duct and is independent of flow rate), all metered flow through 100 channels, and properties evaluated at $\bar{T}_g$. The measured signal exceeds the monolith-only prediction by 1.4 to 2.8 times at high flow (2.74 against 0.99 mbar at 15.3 sL min^{-1}), falls below it at low flow, and goes negative at the lowest (-0.02 against 0.20 mbar at 4.5 sL min^{-1}). This is a shortcoming of the zero offset of the selected transducer whose ± 0.2 mbar accuracy floor, 0.1% of its ± 200 mbar span, is the same order as the entire expected monolith drop of 0.20 to 1.0 mbar.

Pressure drop is the quantity by which the volumetric-receiver literature diagnoses honeycomb flow behaviour. Absorbers whose pressure loss is linear in velocity tend inherently towards flow instability [27], while high-conductivity materials with a predominantly quadratic characteristic are the stable ones [28]. Instability becomes possible when one pressure-drop level maps to more than one outlet temperature, letting a region drift away from the operating point and fail locally if it exceeds the material limit [26]. Ávila-Marín's review extends the criterion to highly porous honeycombs specifically [1], and permeability and Forchheimer coefficients from pressure-drop data are the standard inputs to homogeneous equivalent models [13]. Recent unheated monolith measurements place a honeycomb on the linear, instability-prone side of that axis. The first experimental validation of a dual-cell-density pressure-drop model finds a linear relationship between Reynolds number and pressure drop in every configuration tested, while progressively underestimating the drop as Re rises through entrance and

exit effects [45]. Those effects are large enough at channel Re of 50 to 320 to require an explicit correction factor, strongly dependent on spacing and increased by misalignment [44]. Our transducer, spanning monolith plus plumbing, cannot separate the absorber's characteristic from those comparable losses. However, we still retain and utilize these measurements herein because i) the fix is inexpensive, and ii) section 5.3 shows this to be the measurement that would discriminate between the two candidate mechanisms above. Therefore, supplementary section S8 gives the per-run comparison.

5. Two consequences for reduced-order receiver modelling

5.1 Against validating receiver models on a gas outlet temperature and one or two solid temperatures

The standard validation dataset for a volumetric receiver model is an outlet gas temperature together with one or two embedded solid temperatures. Almost every campaign surveyed above reports these two quantities — outlet air temperature and efficiency, with front-face temperature by pyrometer or infrared camera where available [4,5,6,7,8,9]. The convention is explicit at review level: He and co-workers name outlet air temperature and thermal efficiency as the two key indicators for solar receivers, noting that a non-uniform radiation distribution produces a non-uniform air temperature distribution requiring more thermocouples than campaigns typically deploy [46]. Modelling studies calibrate and validate against those same quantities [10,11,12,13], and recent practice has not moved past it [29,31,34]. This campaign contains the same measurements several times over, permitting a direct test of how much such a dataset constrains. The answer is very little, far less than the apparent quality of the fit suggests.

The gas outlet temperature is nearly blind to the internal field. At 456 kW m^{-2} T_3 varies by 26 K across the flow range while the front-face wall varies by 264 K; at 304 kW m^{-2} the ratio is 51 K against 331 K. Worse, T_3 's flow slope changes sign between configurations, so the qualitative behaviour a model must match is inconsistent across the campaign.

Which solid probe is called the solid temperature changes every identified coefficient. Recomputing the entire analysis, using the same fifteen runs, reduction, and fitting procedure but varying only which thermocouple defines T_s , gives Figure 6a:

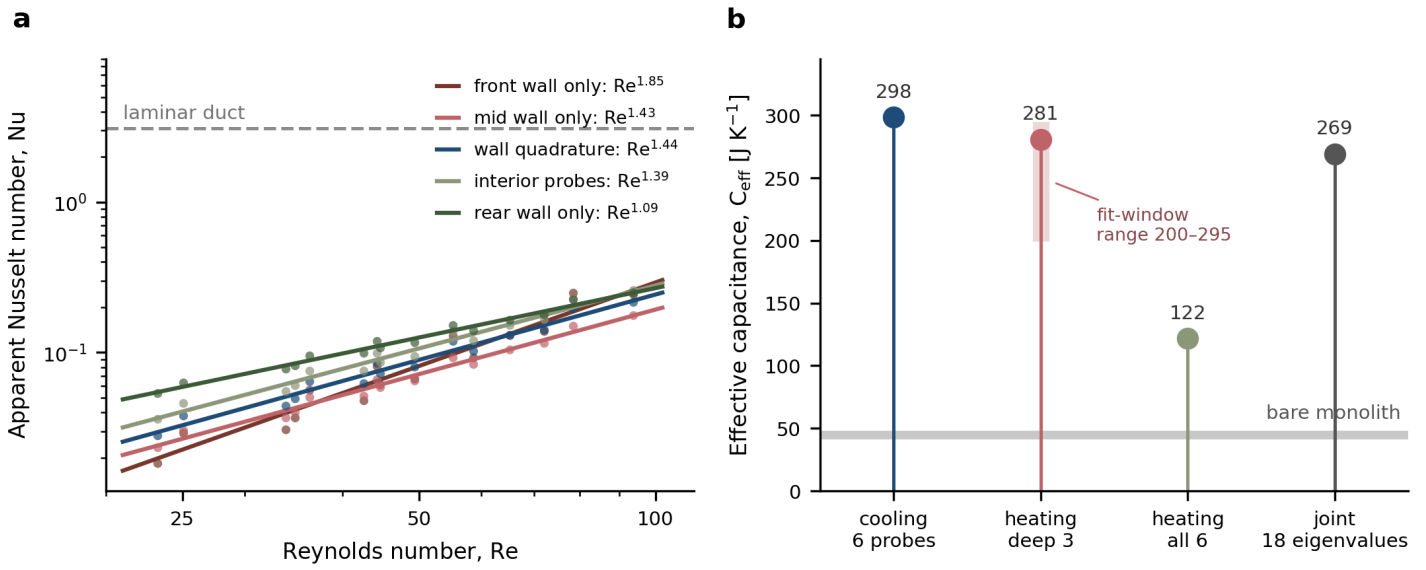


Figure 7. (a) The apparent Nusselt correlation recomputed against five different solid reference probes from the same fifteen runs: the prefactor spans a factor of 29.7 and the exponent 1.09 to 1.85. (b) The effective capacitance from the same eighteen transients under four defensible analysis choices, with the fit-window range and the bare monolith capacitance marked.

Solid reference	ε range	Nu prefactor	Nu exponent	ε^* (456/304/256)
front wall only (T8)	0.454 – 0.851	5.91×10^{-5}	1.847	0.605 / 0.591 / 0.572
mid wall only (T12)	0.520 – 0.710	2.70×10^{-4}	1.428	0.604 / 0.590 / 0.570
wall quadrature (adopted)	0.573 – 0.781	3.14×10^{-4}	1.444	0.666 / 0.655 / 0.642
wall + interior mean	0.608 – 0.803	3.75×10^{-4}	1.421	0.691 / 0.682 / 0.671
interior probes (T9, T10)	0.648 – 0.826	4.56×10^{-4}	1.394	0.719 / 0.711 / 0.702
rear wall only (T11)	0.771 – 0.823	1.76×10^{-3}	1.092	0.787 / 0.787 / 0.791

Table 5. The apparent Nusselt correlation and the inversion marker recomputed against five candidate solid reference temperatures from the same fifteen runs.

The prefactor spans a factor of 29.7, the exponent ranges from 1.09 to 1.85, and the inversion threshold spans 0.57 to 0.79 (Table 5). Every row is a defensible unremarked choice, and every row is a different receiver. Interior and wall probes at the same depth differ by 21–55 K at 107 mm and 21–27 K at 58 mm, with the gaps growing with flow, so no single solid temperature represents the section.

Although the same sensitivity is documented channel-side [17,19], at assembly scale the effect is a factor of twenty-nine, not tens of percent. The transient identification is equally under-determined: the same eighteen transients yield C_{eff} of 122, 269, 275 or 281 J K⁻¹ depending on probe subset and fit window. The window alone spans 200–295 J K⁻¹, with the coefficient of determination increasing as the estimate degrades (Figure 7b). This factor of 2.3 is the width of the identifiability basin, not measurement scatter.

Therefore, agreement between a receiver model and a gas outlet temperature plus one or two solid temperatures is not evidence of correct internal physics; the identified coefficient reflects sensor placement rather than the receiver and will not transfer to a different geometry, scale or operating envelope. This offers a deflationary explanation for a two-decade puzzle: most tested volumetric absorbers underperformed predictions [1,3], four established models cluster within 25 K of each other and 80–100 K from experiment with mean-air-temperature measurement named a principal suspect [10], and the idealized equilibrium model is optimistic by over twenty efficiency points against practice [2]. If the calibration target does not constrain the internal field, models can be simultaneously well fitted and structurally wrong.

Two external results sharpen this. Iding and co-workers reproduce the front temperature of the 1080-cup Jülich receiver to 6.78 K RMS and the behind-cup air temperature to 7.51 K, on the two-quantity dataset described above. However, they report that this per-cup performance does not carry to the array [30]: the field resolves the irradiated surface to a few kelvin, but not the depth direction in which the identified coefficients are defined. Conversely, Song and co-workers resolve the interfacial coefficient in porous media at low Reynolds number by measuring solid and gas temperatures simultaneously at each of a series of axial stations and solving the discretized gas energy equation directly, rather than inverting against a temperature field. They note that this inversion approach is ill-posed [33]. Their term for co-located temperatures is data integrity, and the present results quantify the error from choosing the solid reference instead.

We recommend that reported receiver-model validations state explicitly which solid temperature defines the reference. However, there are two cases to consider regarding the identified coefficient's sensitivity to that choice. First, where the instrumentation permits, that sensitivity should be reported. Second, where it does not permit, the coefficient should be reported as conditional rather than as a receiver property. This is not an argument against reduced-order models but that the standard validation dataset cannot discriminate among them — the discriminating measurements are different ones.

5.2 Against uniform-temperature or uniform-flux heat transfer coefficients in zero- and one-dimensional receiver models

Zero- and one-dimensional receiver models close the gas–solid coupling with a coefficient from a correlation derived for a duct with uniform wall temperature or uniform wall flux: a Churchill cross-flow correlation in the inlet region [10], volumetric forms $Nu_{lv} = a Re^b Pr^c$ [13], the Fu and Kuwahara forms [15], or Graetz-problem Nusselt and Sherwood numbers at constant wall

temperature [21]. That closure is inadmissible, and no model is needed to show it. Every such model reduces to the single-stream form

$$\dot{m}c_p \frac{dT_g}{dz} = h(z) P [T_s(z) - T_g(z)],$$

and if h is fixed in position, as a Nusselt number on unchanging local geometry implies, integration gives the transfer-unit count $N = (1/\dot{m}c_p) \int h P dz$, whose flow dependence is that of $1/\dot{m}$ alone.

Section 4.2 evaluates this on measured properties as -1.000 but measures $+0.341 \pm 0.040$ instead, so the sign is wrong. Four local repairs were tested and each fails: a developing-flow correlation [17,24] scales h no faster than $Re^{0.5}$ and worsens the disagreement; solid-side resistance is ruled out by $Bi \leq 3.4 \times 10^{-5}$; axial gas conduction by $Re \cdot Pr \cdot L/D_h \geq 1460$; and a uniform imposed flux is inadmissible because N_{rc} of 1.50 to 5.66 means the solid redistributes absorbed energy axially by radiation [14,15,19].

A Reynolds exponent above unity might seem to disqualify this as a heat-transfer correlation, but such exponents are typical of interfacial coefficients measured in porous media at these Re . Song and co-workers compile eight correlations for $Re \leq O(10^2)$ with exponents from 0.2 to 1.97, adding their own, $Nu = 0.001 Re^{1.66} Pr^{1/3}$, $r^2 = 0.93$ over $13 \leq Re \leq 133$ from fifty-four tests [33]; our 1.444 over $Re = 23\text{--}94$ sits inside that range. The comparison is one of exponent and magnitude only — their coefficient is volumetric and particle-diameter-referenced in a packed bed, ours apparent and wall-referenced in a honeycomb — showing that a superlinear exponent recurs when this coefficient is measured rather than assumed at low Re , not that the values should agree. This is not a competing channel correlation: the channel-scale work correctly answers its own question, while ours is an assembly-scale quantity in the same non-dimensional form, and the gap between the two is a measure of how little of the structure participates under the assumptions set out in section 3.1.

That inversion assumes spatially uniform h , excluding only a constant-Nusselt closure, not every fixed $h(z)$. To test the broader claim, $h(z)$ is fitted across three families, each piecewise linear and non-negative on evenly spaced nodes and fitted by multi-start least squares in $\log h$, asking whether one flow-independent profile reproduces the fifteen measured outlet temperatures. First, a single dimensional $h(z)$ is shared by all runs. Second, a separate five-node profile is fitted per irradiance configuration, so that a conductance differing with lamp setting or temperature level is not held against the hypothesis. Third, a shared Nusselt profile, $h_i(z) = Nu(z) k_i/D_h$, is fitted with each run's own conductivity; this is the direct test of the constant-Nusselt class.

None succeeds; the discriminating statistic is the ordering of the residual in flow, not its magnitude. For the shared conductance profile, the RMS outlet residual is 64 K with two nodes and saturates at 55, 54 and 54 K with three, five and seven, so the residual is not limited by the number of free nodes; residuals correlate with $\ln Re$ at $r = -0.95$, falling 126–148 K per e-fold, from about +80 K at the lowest flow of each configuration to –70 K at the highest. As with the shared conductance profile, the shared Nusselt profile behaves identically, at 56 and 55 K with three and five nodes. The per-configuration profile lowers the residual in one group and raises it in another (36, 68, 52 K at 256, 304, 456 kW m⁻²). However, it sharpens the ordering to $r = -0.996, -0.982, -0.997$: a flow-independent profile fails monotonically even within a single lamp configuration. A fixed conductance over-predicts

outlet temperature at low flow and under-predicts it at high flow in every family, a trend the ± 25 K outlet-bias band cannot absorb. However, two limits apply. First, the test presumes the single-stream form, excluding only flow-independent conductance within it. Second, it inherits the wall reconstruction's bounded but unmeasured extrapolations. Neither affects the sign or ordering. All three families, fitted nodes and per-run residuals are archived under `fixed_profile_test`.

Taken at face value, the measurement says participating gas–solid contact grows with Reynolds number. Two confounders would produce the same signature without any change in participation: an outlet-probe bias varying systematically with flow, and a departure from thermal stationarity varying with run duration and hence with flow. This campaign bounds neither, and the first two measurements of section 5.3 are directed at them. Read physically, what a single-stream model fixes but this receiver need not is the distribution of flow among channels. Two mechanisms, advanced as leading hypotheses rather than identified causes, produce the required sign. Viscosity-driven redistribution at common pressure drop operates as follows: hot core channels choke relative to cooler perimeter channels, so as total flow rises the field cools, the viscosity contrast weakens, and the participating fraction grows. This predicts that transfer units and the wall-to-interior deficit both rise with Re , with a front-to-mid wall crossing, all observed in sections 4.1–4.3, matching the coupling reported for porous SiC absorbers where non-uniform hydrodynamics are significantly influenced by increased viscosity at the hot spot [9]. Peripheral bypass: a leakage path through the housing clearance whose share falls as total flow rises, being cooler with resistance a weaker function of temperature, gives the same sign while inflating a nominal Reynolds number. The two are indistinguishable from temperature data alone, both require a flow-distribution degree of freedom absent from a single-stream model, and differ in that redistribution conserves the flow path while bypass short-circuits it. Section 5.3 measures this distinction.

The first mechanism is the flow-instability criterion carried by the volumetric-receiver literature for three decades (section 4.7 [26,27,28]), restated here as continuous rather than catastrophic. The linear regime is viscosity-dominated, using the viscosity of the heated gas, so the contrast driving runaway overheating in an unstable absorber must, in a stable one, still redistribute flow reversibly, weakening as total flow rises. The measured sign is the same, and the criterion attaches specifically to highly porous honeycombs [1]. The parallel-channel literature bears on it structurally rather than quantitatively, since both results are for flow boiling and are therefore cited for structure, not for transfer of values. Lateral channel-to-channel conductance alleviates maldistribution, with a single eigenvalue setting its onset [39], consistent with radial heat transfer reducing instability probability in porous absorbers [28]. However, a maldistributed state can be thermodynamically favoured over an even one [40]. The radiation–conduction number measured here is such a lateral conductance, so transfer units rising with Reynolds number occur in the presence of, not the absence of, a stabilizing mechanism.

The remedy is a multi-stream or distributed formulation with a pressure-coupled flow split, not a better Nusselt correlation. This is preceded in pseudochannel models built because single-channel models cannot predict accurately under flow maldistribution [35], in flow-distribution models for dual-cell-density monoliths [36] and dual-zone packed beds [43], and in solar-field models at panel and module scale [34,37]. A dual-cell-density monolith with a Darcy momentum sink, isothermal,

gives a flow fraction depending only on relative permeability and core size, not flow rate [36]. Flow rate cancels from the equal-pressure-drop condition when every path is linear in velocity with one shared viscosity. Adding an inertial term via the Ergun equation makes the split Reynolds-dependent [43]. Therefore, a flow-dependent split needs an inertial contribution or a path-to-path viscosity difference, and both are present: the heated absorber carries a non-uniform viscosity field [9], and the viscous-to-inertial transition is the axis of the stability criterion itself [27,28]. Buoyancy supplies a third such agent in molten-salt panels [37], and in all three the split is most differentiated at low flow. Flow-dependent participation is a general property of heated parallel-path arrays. Supplementary section S7 develops the comparison.

The scope claimed is precise: not that no continuum model can reproduce these data, nor an impossibility over all parameters. Rather, a single-stream formulation whose volumetric conductance is a fixed function of axial position cannot produce a transfer-unit count increasing with flow, and this receiver's does. Any such model, however calibrated, will misrepresent the flow response. Since such models are typically calibrated against the dataset section 5.1 shows uninformative, the misrepresentation can survive an apparently successful validation. Although this does not contradict the catalysis literature's continuum reductions, it delimits them: a pseudo-continuous model reproduces three-dimensional CFD to better than 0.5 K assuming flow equally distributed over all channels [21], and effective radial conductivities are obtained with the gas static [20]. This is reasonable for a reactor with a heated wall and metered uniform inflow. What fails here is not the reduction but its use under a concentrated, spilling beam with an insulated housing and a flow split the boundary conditions do not control.

5.3 What would resolve it

Three measurements would settle the questions left open, ranked by value per unit cost. Most valuable is a differential pressure measurement across the monolith faces alone with a ± 2 to ± 5 mbar transducer. The two candidate mechanisms of section 5.2 have different pressure-drop signatures at the same total flow, since redistribution conserves the flow path while bypass short-circuits it. A pressure-drop series across the flow range would discriminate them, preferably run with a sealed-clearance control to remove the bypass path. A single point would not, since the two mechanisms differ in the flow dependence of the characteristic rather than in its value at one flow. The expected signal is 0.20 to 1.0 mbar, resolvable to a few percent with an appropriately ranged transducer. It would also connect this receiver to the permeability and Forchheimer characterization that the homogeneous-equivalent modelling literature depends on [13]; these are the linear and quadratic coefficients relating pressure drop to flow rate. In particular, it would place the absorber on one side or the other of the linear-versus-quadratic flow-stability criterion [26,27,28]. Its practicability at this scale is shown by a calorimetric test bench for pressurised gas receivers carrying absolute pressure sensors before and after the absorber alongside the thermocouples [32]. No measurement of channel-to-channel flow split in an irradiated honeycomb absorber appears in the recent literature. Broeske and co-workers state that individual mass-flow measurements for each of their sixteen absorber modules were impossible for reasons of space, so efficiencies were evaluated through radiative energy balance rather than air enthalpy flow [29]; second, the nearest published measurements isolate the effect in surrogates, varying air velocities in a two-tube fluidised-particle

receiver at ambient temperature to impose controlled thermal heterogeneity and observe the resulting particle mass-flow split [38]. The measurement proposed here is not a replication.

Second is an independent determination of outlet enthalpy, by mixing calorimeter or shielded aspirated probe. This would break the T3 dependence propagating into effectiveness, transfer units, the Nusselt correlation, the inversion threshold, efficiency, and the identified constants. It would address the mean-air-temperature difficulty named as a principal source of the field's model-experiment gap [10]. Third is a radial thermocouple traverse at one depth to resolve the core-to-perimeter solid profile, since section 5.1 identifies a single solid temperature per depth as the binding limitation on identifiability. Two radial positions at one depth would test the redistribution mechanism directly and replace the reference-choice sensitivity of Figure 6a with a measured profile.

5.4 Limitations

Several limitations bound what has been established. The isothermal-wall ϵ -NTU relation, in which NTU denotes the number of transfer units relative to the stream's heat-capacity rate, approximates a wall spanning up to 321 K. The bias this introduces is quantified: profile-corrected transfer units exceed identity values by 1.5% to 8.7%, shifting the exponent from +0.389 to +0.341. That correction inherits the wall reconstruction, whose front 11 mm and rear 30 mm are extrapolated rather than measured. The rear extrapolation is bounded by the outlet temperature but not determined by it. The wall probes' installation error, comprising contact resistance and cavity radiation, is unquantified. The outlet-probe systematic bias is bounded by declaration rather than measurement. However, there are two aspects to be considered before accepting that bound. First, the bound applies only against a uniform offset. Second, a bias varying with flow is unbounded and would act directly on the exponents carrying the paper's argument. Thermal stationarity at the end of each heating window is assumed, not demonstrated. The primary transient identification rests on one degree of freedom. The inversion threshold at 256 kW m^{-2} is sensitive to the crossing estimator at a level comparable to its flux dependence, and that group contains a single negative point. Flux-map uncertainty is characterised by its discrepancy with the energy closure rather than propagated from the instrument chain. Pressure drop is unusable as installed.

Two thermal models are solved: the profile-corrected transfer-unit count and the fixed-conductance falsification of section 4.2. Both integrate a single-stream gas energy equation through the measured wall profile. This is deliberately a model of the class the paper argues against, used to test that class rather than correct any measurement. Therefore, nothing here resolves the internal flow field, and no reported measurement is corrected using a model of it.

6. Conclusions

Gas-solid exchange in this structured SiC receiver is limited at the assembly scale rather than in the channel. The apparent Nusselt number referenced to the length-averaged wall temperature follows $Nu_{app} \propto Re_{nom}^{1.472 \pm 0.011}$ with $r^2 = 0.9993$ under a common-slope, per-configuration-prefactor fit over Reynolds numbers of 23 to 94, prefactors 2.55 to 3.24×10^{-4} ; the pooled form gives 1.444 ± 0.072 at $r^2 = 0.968$. Either way this is 14.3 to 110 below the fully developed laminar square-

duct value $Nu_{H2} = 3.091$ and far below channel-level monolith-literature values, with a web Biot number of at most 3.4×10^{-5} excluding a through-web conduction resistance as the limiting one, though not every solid-side or assembly-scale resistance. The front-to-mid side-wall crossing collapses onto a single dimensionless marker under the adopted wall-averaging convention, the mid-depth wall overtaking the front face once the gas has recovered about two thirds of the available wall-to-ambient temperature difference, at $\epsilon^* = 0.666, 0.655$ and 0.642 ; the criterion is robust in form, but its weak flux dependence, comparable to the estimator systematic, is unresolved. On this plain, ungraded honeycomb the side-wall signature of volumetric behaviour is thus at least as much an operating condition as a design property. The apparent wall-to-interior disequilibrium grows linearly with flow at both depths, with a flux-invariant slope of $(8.54 \pm 0.36) \times 10^{-4}$ per unit Re at 107 mm and roughly a third of that at 58 mm — a smooth function of depth and flow, not a threshold phenomenon — giving the two-equation modelling literature a measured number to check against.

The receiver's transient is set by six to seven times the absorber's own thermal mass. The slow eigenvalue is linear in the gas advective conductance, giving $C_{eff} = 275 \pm 38 \text{ J K}^{-1}$ from cooling and $269 \pm 30 \text{ J K}^{-1}$ from a joint fit to all eighteen eigenvalues against a bare monolith capacitance of 42 to 47 J K^{-1} , with loss conductance bracketed at 0.080 to 0.114 W K^{-1} . All fifteen heating transients rescaled on this eigenvalue collapse to a wall half-rise of $t^* = 0.205$ within 16.8%, though not a one-state response since the wall retains a faster component the lumped model omits.

A transfer-unit count increasing with flow rules out a widely used reduced-order model class: integrated from the measured axial wall profile, it scales as $Re_{nom}^{+0.341 \pm 0.040}$, opposite in sign to the $Re_{nom}^{-1.00}$ required by any single-stream formulation with position-fixed volumetric conductance, since channels are fully developed over at least 96% of their length, giving -1.000 with developing-flow closure only widening the gap. Neither uniform, shared, nor separately fitted two-to-seven-node conductance/Nusselt profiles reproduce the outlet-temperature flow response. An empirical $h \propto Re^{1.34}$ reproduces the exponent by construction but lacks mechanism or predictive content beyond the fitted range; a flow-distribution degree of freedom would supply one, but this campaign cannot discriminate it from flow-dependent probe bias or non-stationarity.

Validation against a gas outlet temperature and one or two solid temperatures does not constrain receiver physics: from the same fifteen runs, changing the reference solid thermocouple moves the identified Nusselt prefactor by a factor of 29.7, the exponent from 1.09 to 1.85, and the inversion threshold from 0.57 to 0.79; changing probe subset and fit window moves identified capacitance from 122 to 281 J K^{-1} . At the highest flux the outlet gas spans 26 K across the flow range while the front-face solid spans 264 K.

Delivered aperture irradiance is not determined to a value. Radiometric flux mapping and steady energy closure disagree by -1 to $+13\%$, $+15$ to $+32\%$, and -22 to -8% across the three configurations, with flow-dependent discrepancy precluding a uniform correction; a Gardon gauge over 499 mm^2 cannot resolve flux from a 361 mm^2 face in a focused spot, giving a geometric bias of 1.04 to 1.28. Reported spans are 451–517, 304–402, and 201–256 kW m^{-2} . No temperature-based result depends on this choice.

The discriminating measurement is a properly ranged differential pressure across the monolith faces, taken as a series across the flow range: with an accuracy floor of ± 0.2 mbar against an expected 0.20 to 1.0 mbar signal, the installed transducer cannot constrain flow distribution, whereas a ± 2 to ± 5 mbar device — with a sealed-clearance control run — would separate viscosity-driven redistribution from peripheral bypass.

Data availability and reproducibility

The raw data for the experimental campaign described here in are available at the public repository [give github](#). Every derived table, uncertainty estimate and figure input in the main text and in the supplementary material is produced by one script, `receiver_reduction.py`; the figures are then drawn from those outputs by a second script, `make_figures.py`, which writes Figures 3 to 7. Both scripts are (random number) seeded, so a rerun on the software stack given in section 3.4 can reproduce all the outputs, except of the multi-start optimizer of section 5.2 whose last digits may differ across BLAS builds. The same stands for the three tables in the main text along side tables S3-5 in supplementary. The supplementary material and its accompanying README document the reproduction commands and the interim generated output files in full.

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